

■ EXAMPLE 11.5

We will revisit Example 11.2, but assume that the initial state is 0:

$$\begin{aligned}x_{k+1} &= x_k + w_k \\x_0 &= 0 \\y_k &= x_k + v_k \\z_k &= x_k\end{aligned}\tag{11.140}$$

From Equation (11.131) we can find the *a priori* steady-state filter that bounds the infinity-norm of the transfer function from e to $\tilde{z}$ by $1/\sqrt{\theta}$. (Recall that $e_k = [w_k \ v_k]^T$.) The algebraic Riccati equation associated with this problem is given by

$$\begin{aligned}P &= 1 + P - P(1 + P)^{-1}P + P(1/\theta + P)^{-1}P \\&= 1 + P - \frac{P^2}{1 + P} + \frac{P^2}{1/\theta + P}\end{aligned}\tag{11.141}$$

Solving the above for P we obtain

$$P = \frac{-\theta - 1 \pm \sqrt{\theta^2 - 6\theta + 5}}{2(2\theta - 1)}\tag{11.142}$$

In order for the solution of this equation to solve the H_∞ filtering problem, we must have $P > 0$. The only solution for which $P > 0$ is when $0 \leq \theta < 1/2$ and when we use the negative sign in the above solution.⁵ If we choose $\theta = 1/10$ then $P = 2$. The gain of the *a priori* filter is then computed from Equation (11.131) as

$$\begin{aligned}K &= P(1 + P)^{-1} \\&= 2/3\end{aligned}\tag{11.143}$$

Note that the P value that is obtained for $\theta = 1/10$ does not match Example 11.2, but K does match. The H_∞ filter equation is computed from Equation (11.131) as

$$\begin{aligned}\hat{x}_{k+1} &= \hat{x}_k + K(y_k - \hat{x}_k) \\&= \hat{x}_k + (2/3)(y_k - \hat{x}_k)\end{aligned}\tag{11.144}$$

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11.6 SUMMARY

In this chapter, we have presented a couple of different approaches to H_∞ estimation, also called minimax estimation. H_∞ filtering minimizes the worst-case

⁵Note that Example 11.2 showed that this problem has a solution for $0 \leq \theta < 1$, which indicates that the game theory approach to H_∞ filtering may be more general than the transfer function approach.

estimation error and is thus more robust than Kalman filtering, which minimizes the RMS estimation error. H_∞ filtering has sometimes been criticized for being too pessimistic in its assumption about the noise processes that impinge on the system and measurement equations. After all, H_∞ estimation assumes that the noise is worst case, thus attributing a degree of perversity to the noise that intuitively seems unrealistic. This has led to mixed Kalman/ H_∞ estimation techniques, which we will discuss in Chapter 12.

Research in H_∞ estimation began in the 1980s. During that decade, some work was directed toward the design of minimax state estimators for systems corrupted by random noise whose covariances were within known bounds [Poo81, Dar84, Ver84]. This was a first step toward H_∞ filtering, although it still assumed that the noise was characterized by statistical measurements. The earliest work that could pass for what we now call H_∞ filtering was probably published by Mike Grimble [Gri88]. However, unlike the presentation in this chapter, he used a frequency domain approach. He designed a state estimator such that the frequency response from the noise to the estimation error had a user-defined upper bound.

Some early tutorials on H_∞ filtering can be found in [Gri91b, Sha92]. A polynomial systems approach to H_∞ filtering is presented in [Gri90]. Nonlinear H_∞ filtering is discussed in [Rei99], where a stable state estimator with a bounded infinity-norm is derived. System identification using H_∞ methods is discussed in [Sto94, Tse94, Bai95, Did95, Pan96].

The effectiveness of the H_∞ filter can be highly sensitive to the weighting functions [e.g., S_k , P_0 , Q_k , and R_k in Equation (11.36), and θ in the performance bound]. This sometimes makes H_∞ filter design more sensitive than Kalman filter design (which is ironic, considering the higher degree of robustness in H_∞ filtering). The advantages of H_∞ estimation over Kalman filtering can be summarized as follows.

1. H_∞ filtering provides a rigorous method for dealing with systems that have model uncertainty.
2. H_∞ filtering provides a natural way to limit the frequency response of the estimator.

The disadvantages of H_∞ filtering compared to Kalman filtering can be summarized as follows.

1. The filter performance is more sensitive to the design parameters.
2. The theory underlying H_∞ filtering is more abstract and complicated.

The types of applications where H_∞ filtering may be preferred over Kalman filtering could include the following.

1. Systems in which stability margins must be guaranteed, or worst-case estimation performance is a primary consideration (rather than RMS estimation performance) [Sim96].
2. Systems in which the model changes unpredictably, and identification and gain scheduling are too complex or time-consuming.
3. Systems in which the model is not well known.

Work by Babak Hassibi, Ali Sayed, and Thomas Kailath involves the solution of state estimation problems within the context of Krein spaces (as opposed to the usual Hilbert space approach). This provides a general framework for both Kalman and H_∞ filtering (along with other types of filtering), and is discussed in some of their papers [Has96a, Has96b] and books [Has99, Kai00].

PROBLEMS

Written exercises

11.1 Show that $(I + A)^{-1}A = A(I + A)^{-1}$.

11.2 Consider a scalar system with $F = H = 1$ and with process noise and measurement noise variances Q and R . Suppose a state estimator of the form

$$\hat{x}_{k+1}^- = \hat{x}_k^- + K(y_k - \hat{x}_k^-)$$

is used to estimate the state, where K is a general estimator gain.

- Find the optimal gain K if $R = 2Q$. Call this gain K_0 . What is the resulting steady-state *a priori* estimation-error variance?
- Suppose that $R = 0$. What is the optimal steady-state *a priori* estimation-error variance? What is the (suboptimal) steady-state *a priori* estimation-error variance if K_0 is used in the estimator? Repeat for $R = Q$ and $R = 5Q$.

11.3 Consider a scalar system with $F = H = 1$ and with process noise and measurement noise variances Q and $R = 2Q$. A Kalman filter is designed to estimate the state, but (unknown to the engineer) the process noise has a mean of $\bar{w}$.

- What is the steady-state value of the mean of the *a priori* estimation error?
- Introduce a new state-vector element that is equal to $\bar{w}$. Augment the new state-vector element to the original system so that a Kalman filter can be used to estimate both the original state element and the new state element. Find an analytical solution to the steady-state *a priori* estimation-error covariance for the augmented system.

11.4 Suppose that a Kalman filter is designed to estimate the state of a scalar system. The assumed system is given as

$$\begin{aligned}x_{k+1} &= Fx_k + w_k \\ y_k &= Hx_k + v_k\end{aligned}$$

where $w_k \sim (0, Q)$ and $v_k \sim (0, R)$ are uncorrelated zero-mean white noise processes. The actual system matrix is $\tilde{F} = F + \Delta F$.

- Under what conditions is the mean of the steady-state value of the *a priori* state estimation error equal to zero?
- What is the steady-state value of the *a priori* estimation-error variance P ? How much larger is P because of the modeling error ΔF ?

11.5 Find the stationary point of $(x_1^2 + x_1x_2 + x_2x_3)$ subject to the constraint $(x_1 + x_2 = 4)$ [Moo00].

11.6 Maximize $(14x - x^2 + 6y - y^2 + 7)$ subject to the constraints $(x + y \leq 2)$ and $(x + 2y \leq 3)$ [Lue84].

11.7 Consider the system

$$\begin{aligned}x_k &= \frac{1}{2}x_{k-1} + w_{k-1} \\ y_k &= x_k + v_k\end{aligned}$$

Note that this is the system model for the radiation system described in Problem 5.1.

- a) Find the steady-state value of P_k for the H_∞ filter, using a variable θ and $L = R = Q = S = 1$.
- b) Find the bound on θ such that the steady-state H_∞ filter exists.

11.8 Suppose that you use a continuous-time H_∞ filter to estimate a constant on the basis of noisy measurements. The measurement noise is zero-mean and white with a covariance of R . Find the H_∞ estimator gain as a function of P_0 , R , θ , and time. What is the limit of the estimator gain as $t \rightarrow \infty$? What is the maximum value of θ such that the H_∞ estimation problem has a solution? How does the value of θ influence the estimator gain?

11.9 Prove that $\mathcal{H}$ and $\tilde{\mathcal{H}}$ in Equations (11.134) and (11.137) are symplectic.

11.10 Prove that the solution of the *a posteriori* H_∞ Riccati equation given in Equation (11.132) with $\theta = 0$ is equivalent to the solution of the steady-state *a priori* Kalman filter Riccati equation with $R = I$ and $Q = I$.

11.11 Prove that Σ in Equation (11.132) with $\theta = 0$ is equivalent to the solution of the steady-state *a posteriori* Kalman filter Riccati equation with $R = I$ and $Q = I$.

11.12 Find the *a posteriori* steady-state H_∞ filter for Example 11.5 when $\theta = 1/10$. Verify that the *a priori* and *a posteriori* Riccati equation solutions satisfy Equation (11.133).

11.13 Find all possible solutions P to the *a priori* H_∞ filtering problem for Example 11.5 when $\theta = 0$. Next use Equation (11.139) to find the P solution. Repeat for $\theta = 1/10$. [Note that Equation (11.139) gives a negative solution for P and therefore cannot be used.]

Computer exercises

11.14 Generate the time-varying solution to P_k for Problem 11.7 with $P_0 = 1$. What is the largest value of θ for which Equation (11.90) will be satisfied for all k up to and including $k = 20$? Answer to the nearest 0.01. Repeat for $k = 10$, $k = 5$, and $k = 1$.

11.15 Consider the vehicle navigation problem described in Example 7.12. Design a Kalman filter and an H_∞ filter to estimate the states of the system. Use the

following parameters.

$$\begin{aligned}
 T &= 3 \\
 u_k &= 1 \\
 Q &= \text{diag}(4, 4, 1, 1) \\
 R &= \text{diag}(900, 900) \\
 \text{heading angle} &= 0.9\pi \\
 x(0) &= \hat{x}(0) = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}^T
 \end{aligned}$$

Simulate the system and the filters for 300 seconds. In the H_∞ filter use $S = L = I$ and $\theta = 0.0005$.

- a) Plot the position estimation errors for the Kalman and H_∞ filters. What are the RMS position estimation errors for the two filters?
- b) Now suppose that unknown to the filter designer, $u_k = 2$. Plot the position estimation errors for the Kalman and H_∞ filters. What are the RMS position estimation errors for the two filters?
- c) What are the closed loop estimator eigenvalues for the Kalman and H_∞ filters? Do their relative magnitudes agree with your intuition?
- d) Use MATLAB's DARE function to find the largest θ for which a steady-state solution exists to the H_∞ DARE. Answer to the nearest 0.0001. How well does the H_∞ filter work for this value of θ ? What are the closed-loop eigenvalues of the H_∞ filter for this value of θ ?

